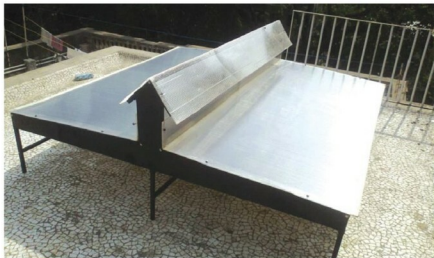


# Solar Conduction Dryer: A Boon to Indian Farmers

A start-up called Science for Society has designed and developed an innovative solar conduction dryer (SCD), with which it aims to improve the economic situation of Indian farmers, by avoiding crop loss. EFY got in touch with Vaibhav Tidke of Science for Society to discuss his innovation



Solar conduction dryer

## ABHISHEK MUTHA

Dehydration, a promising but electricity-dependent technology, is commonly used post harvest to increase the shelf life of food produce by almost a year. Across rural India, power cuts for more than eight hours a day are routine, and therefore there is a need for an electricity-free solution. Dehydrated food comprises 45 per cent more nutritional value than open-sun dried food, and hence has about 50 to 200 per cent higher market value. In this context, the solar conduction dryer (SCD) can bring about immense economic improvement in the lives of India's farmers. The SCD is a solar-based device for the dehydration of agro products. It is so innovative that it won a US\$ 60,000 grand prize at the Dell Social Innovation Challenge,

where it competed against various other international projects. This project is also recognised by UNESCO, the National University of Singapore and the government of India.

## Closing the urban-rural economic gap with efficient post-harvest technology

The SCD can be used to dehydrate agro products (fruits and vegetables), fish and meat. In most cases, an innovation is the result of someone looking to solve an existing problem. Talking about what prompted him to design such a solution, Vaibhav Tidke, founder, Science for Society, says, "According to various surveys, almost 30 per cent of the agro products like fruits, vegetables and grains are wasted due to improper storage and lack of affordable and ef-

ficient post-harvest technologies. In a country like India, 70 per cent of the population is directly or indirectly dependent on agriculture, and over 20 per cent of its population is below the poverty line. We cannot afford the luxury of wasting 30 per cent of our agro products," Tidke adds, "Also, suicides among farmers are a major problem in Maharashtra. Wastage of the final agro produce due to lack of proper storage or processing is one of the main reasons attributed to these suicides. This was the scenario that prompted us to develop affordable yet highly efficient technology to prevent the post-harvest losses of agro and fishery produce."

This project is aimed at uplifting farmers, many of whom are in dire financial situations. It is low-cost, easy to operate and therefore can be effortlessly operated even by illiterate farmers. Through the widespread use of this technology, value supplemented food products can also be conceived. The substantial losses typically incurred during transport and storage can be avoided. There is a market for dehydrated goods in both developing and developed countries.

## The principle on which SCD works

The SCD is based on conduction as the major mode of heat transfer, assisted with convection and radiation. Due to this unique technology, the SCD is more efficient, reduces processing time by 40 per cent and costs three times less than conventional solar dryers. These factors reduce capital expenditure con-